Predictive Models of Measles Transmission Under Decreasing Vaccination Coverage

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20 April 2025 - MODULE E

Abstract

This paper attempts to model the possible effects of the decline of vaccination coverage for the MMR vaccine, focusing on Measles, using two different methods - the SEIR-V model, a deterministic model of differential equations proven accurate by the COVID-19 epidemic; and a visual, stochastic simulation made in *Python* where the same parameters were introduced. The goal was to see if one could visualize these equations in action, where individuals move randomly around a grid, and where the parameters introduced are no longer so abstract but become visible. After some testing for different vaccination rates, the equations' models and the simulations predict similar results after enough iterations have been done of the latter. Our simulations show the kind of setup required to simulate measles' extreme rate of transmission, giving another warning of the upcoming dangers of decreasing vaccine coverage worldwide.

Keywords — measles, vaccine, vaccination coverage decline, herd immunity, SEIR-V, outbreaks

1 Introduction and Motivation

Measles is a very contagious virus known to have caused many deaths before the introduction of an effective vaccine in 1963 [IHM24], which has since greatly reduced yearly fatalities. As a result, eradication of measles is considered underway, and the disease is widely regarded as practically eliminated in a majority of the developed world [WHO24].

However, growing anti-vaccine activism, also known as the "anti-vax" movement, has achieved widely spread vaccine distrust through misinformation campaigns, active media presence, and even top cabinet positions in some developed countries; despite the fact that there is no serious scientific debate on their validity. This has lead to a considerable decline in vaccination rates among children and adults alike, which can lead to a reappearance of the disease, as per simulation 3.2.3 . The United States in particular has a vaccination coverage of 92% for the MMR (Measles, Mumps, Rubella) vaccine, against the 97.5% [1] required for herd immunity [WHO23]. As we show later in simulation 3.2.4, a single infected individual may lead to the proliferation of the

[1] While it is usually stated to be 95%, our stochastic models still predict more than half of the

disease in areas of up to 95% vaccination coverage, endangering those who cannot be vaccinated for medical reasons.

The use of the SEIR-V model, a mathematical system to conjecture predictions for the spread of an illness - in the case of the SEIR-V model, in contrast with the SIR model, including vaccination coverage for the population - has the objective of predicting measles' development and consequences with the maximum possible accuracy in a world with decreasing vaccination coverage. These models have been shown to have reliable predictive capabilities since the recent pandemic [Ari+25].

2 Methodology

This project works to ensure a good understanding of the effects of decreasing vaccination rates in different controlled populations, using as an example the ongoing scenario in the USA and other northern-hemisphere countries with measles. This will be accomplished with the use of the stochastic compartmentalized mathematical modelling SEIR-V, and a stochastic simulation based on it, run in the programming language *Python*.

This form of modelling falls into what is known as Agent-Based Modelling (ABM), a method widely used in a great variety of study branches, like in market simulation, traffic and transportation modelling, wildlife modelling, etc. It is characterized by a bottom-up architecture, centred in modelling the individual agents where "each agent individually assesses its situation and makes decisions on the basis of a set of rules. [...] an agent-based model consists of a system of agents and the relationships between them" [Bon02], allowing us to study the general consequences of these individual actions on the whole population.

The parameters defining the agents' actions are usually defined on broad, real-life data; which our project uses from previous studies on the immunity of certain demographics, the rate of the spread of measles, and the effectiveness of its vaccine. The data used is explicitly cited in section 3.1 .

2.1 SEIR-V Model

It has already been mentioned that SIR is compartmentalized, since the agents have individual properties; that is to say, each compartment has their set of defining characteristics that make them distinct from each other. The five compartments in the SEIR-V model are:

1. S: Susceptible - Base individuals, who are not immune nor infected.
2. E: Exposed - Exposed individuals, who have contracted the disease but aren't currently presenting symptoms nor can transmit it yet.
3. I: Infected - Infected individuals, who have the potential to spread the disease.
4. R: Removed - Recovered individuals, who have recovered or died. They stop being part of the simulation, as they are considered immune.
5. V: Vaccinated - Vaccinated individuals, who have immunity from the infection.

susceptible population to contract the illness. While not everyone does contract it, we have gone for a higher, more conservative threshold which we are sure to be correct up to the usual random variation in stochastic modelling.

2.1.1 SIR Model

When using the SEIR-V model, the system can become much more complex than with SIR. Therefore, in order to simplify a bit, we will be using a modified version of the SEIR-V. To understand these changes, an explanation of the base model is needed. The SIR model only has three compartments, and each individual can move through them in one direction, as can be seen in figure 2.1 .

Figure 2.1: A S individual can become I with a β rate, and an I can become R with a γ rate. Figure retrieved from [Tan21] .

The SIR model consists of the following system of differential equations:

$$dS/dt = -\beta SI$$

$$dI/dt = \beta SI - \gamma I$$

$$dR/dt = \gamma I$$

Where beta and gamma are usually constant in these more basic simulations, and $S = S(t)$, $I = I(t)$ and $R = R(t)$, representing the compartment size at each instant in time.

β being the transmissibility rate, it is equal to $k\tau/M$, where k is the mean number of contacts a infected person has per day; τ represents the transmissibility of the disease being modelled; and M being the population size. Therefore the capacity of a S to become I depends on the movement of their population and how infectious that disease is. β being constant means that, as an example, no self-isolation is being done after an outbreak, all compartments take part in society equally.

The recovery rate is $\gamma = 1/d_i$, where d_i represents the duration of the infection in days, which is constant for a specific disease (although it may vary from person to person, SIR maintains a constant, average period of infection).

Some models include Death, where each compartment has its rate for becoming dead, usually μ; however, because death as a direct result of measles occurs only in 1 in 3000 cases, it is assumed that $S(t) + I(t) + R(t) = N$ for all N. Also, because recovery is absolute in measles (immunity is currently, generally believed to last for life [Cen24]) a dead person and an immune person represent the same type of actor in SIR and in our own stochastic simulation.

2.1.2 Modified SEIR-V model

The SEIR-V is a much more realistic model, which allows us to take into account exposed people not being infectious for a period of time before being fully infected, or this same group

developing a resistance to the disease, allowing them to take part in the immune compartment R. A Vaccinated group is also expressed in its system of equations, since a breakthrough could happen, leaving a vaccinated (or recovered) individual susceptible again, normally because of the efficiency of the vaccine not being absolute or because of periodical mutations of the disease (outbreaks).

Since measles is highly contagious, the incubation time is significant and the recovered group and double vaccinated individuals present more often than not a life-long immunity, E cannot become R without going through I, and V and R stays as such for the length of the simulation.

Therefore, this is the system of equations we've designed to represent this modified version of SEIR-V:

$$dS/dt = -\beta SI \tag{1}$$

$$dE/dt = \beta SI - \alpha E \tag{2}$$

$$dI/dt = \alpha E - \gamma I \tag{3}$$

$$dR/dt = \gamma I \tag{4}$$

Notice the new parameter alpha, which is the incubation rate, meaning that $\alpha = 1/d_e$, where d_e are the days an exposed individual starts being infectious, and therefore γ will only take into account the duration of infectiousness. V is not used because it will be shown as a constant in our simulations.

2.1.3 Coding SEIR-V

For the deterministic simulation, Euler's Method will be used. This implies, for example, that if there is a way to determine the slope of $S(t)$ at a moment $t - \Delta t$ (in this case with equation 1), and $S(t - \Delta t)$ is known, then $S(t) = S(t - \Delta t) + dS/d(t - \Delta t) \times \Delta t$ for a sufficiently small Δt in $S(t)$ domain. This process is repeated for every other compartment, so the following will be used setting some initial S_0, E_0, I_0, R_0, and V_0:

$$S(t) = S(t - \Delta t) + dS/d(t - \Delta t) \times \Delta t \tag{5}$$

$$E(t) = E(t - \Delta t) + dE/d(t - \Delta t) \times \Delta t \tag{6}$$

$$I(t) = I(t - \Delta t) + dI/d(t - \Delta t) \times \Delta t \tag{7}$$

$$R(t) = R(t - \Delta t) + dR/d(t - \Delta t) \times \Delta t \tag{8}$$

2.1.4 A deterministic model

All the models explained above are deterministic, meaning that, based on defined initial parameters that act as inputs to a function, the results will be the same every time the simulation is run. In stochastic simulations, there is a random component that slightly alters the results every time the simulation is done.

So, as an example, in a deterministic simulation where 80% of the population will be infected with a disease, and where this same disease has a mortality rate of 20%, we can know for certain that for each run of the code exactly 16% of such population will die, since $0.2 \times 0.8 = 0.16$.

In contrast, with those same rates, a stochastic simulation would have more variance, so the average of deaths would follow a stochastic distribution, such as a normal distribution for a

sufficiently large population. A deterministic model has the property of working exactly the same for any amount of population (if the share of each group remains constant), while stochastic models more accurately represent the chaos in low population environments where a single, random action may very well lead to a completely different outcome.

2.2 Stochastic simulation

While SEIR-V is deterministic, our simulations in *Python* aren't. We have designed them in an attempt to show how the equations actually predict the outcome of an outbreak, even in semi-random environments. The simulation is based on a grid with a certain number of individuals. Each time frame - by default, every 50 milliseconds - all actors move 1 square in a random direction of the 9 available, diagonals allowed, stationary "movement" too. Then, each time two individuals are in the same place, we simulate a "meeting" and see if one infects the other.

Further discussion of how this follows directly from the β value of the SEIR-V equations can be found in section 3.1 , but, to simplify, the grid size and population count define how many meetings there will be, up to random deviation (since movement is random). This represents the same thing as the β value does, and so visually shows what altering this parameter means: A small city, a more densely-packed population; or, contrarily, a wide countryside, and a low population density.

3 Simulations and Results

3.1 Simulation 0

Simulation 0 is a bare-bones version of the other simulations. Here, there is no virus and no vaccination, as shown in figure 3.1 , and the only parameters are:

- Population Count (M)
- Grid Size (N)
- Frames per day (FpD)

While these two values may not be formally found in the differential equations which define the SEIR-V model, they are what makes our β value. β represents de average "meetings" per individual per day, which are usually constant in a differential equation. This is why our model is stochastic, while SEIR-V is deterministic; actor movement is random in the simulation, which implies that the β value oscillates around an average and is not exact. We can see in the video linked in the caption of figure 3.1 [2] how each individual moves, randomly, and affects the chance it meets others.

[2] If it is seen in a printed format, the link is: https://www.youtube.com/watch?v=9ylAjdpqIcY

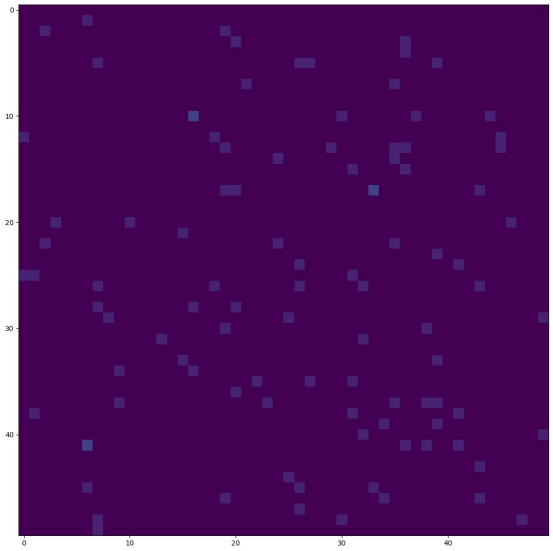
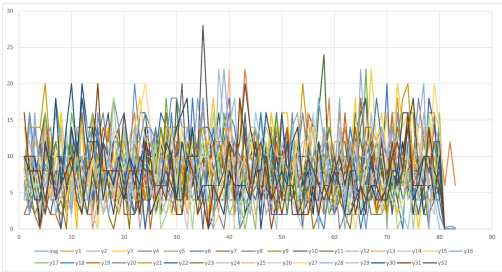


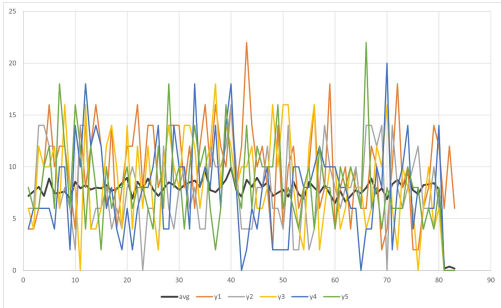
Figure 3.1: A single frame of the video linked in this text. As we can see, the clearer purple squares represent each individual, and the light blue ones represent two individuals on top of each other. The more actors in a single tile, the more contrast the tile has against the background.

This simulation, like the others, updates every 50 milliseconds. Each frame, for each individual, it records any others in its same position, and adds the number of them to a counter b. Then, using the formula $\beta = \frac{b \times FpD}{M \times 100}$ it appends the β value of the frame to a list and resets the b counter. Every 5 seconds, the entire simulation restarts and makes an average of all elements of the list, giving an approximate β value from $5000/50 = 100$ samples. It is important to note that we are assuming a transmissibility of 1, which implies that, later, when we introduce the disease, the beta value will be lowered a bit. This is intended and one should aim to have a beta value slightly higher than real-life data to account for this difference later.

Running this simulation a lot of times and tweaking the parameters above is how we approximate the β value for the disease we are studying. An example of this is figure 3.1 , with parameters $FpD = 10$ (this is constant for all simulations), $N = 50$ and $M = 100$. We can visualize in figure 3.2 some of these β lists in graphic form, and an average of all of their final values, for improved accuracy.



(a) 32 simulations



(b) Average of 5 of them

Figure 3.2: Here we can see, to the left, 32 simulations' β lists in graph form, and, to the right, a simplified graphic, with only 5 simulations, and an average listed. The y-axis is $\beta \times 100$ and x-axis are frames. Because β clearly oscillates around 0.07, we can see that with $N = 50$ and $M = 100$ the disease represented is a bit less contagious than the Covid-19 default strain levels of contagious, which are $\beta = 0.1$ [RWC20].

If one wanted to model a tightly packed city, for example, or the sparse countryside, they would modify the β value in the SEIR-V equations. In our simulations, they'd run Simulation 0 a bunch of times, altering the grid size and population count, to approximate the intended β value, knowing that 10 frames make up a day. This has the perk of allowing an actual visualization of a difference in population density, as can be seen in figure 3.3 .

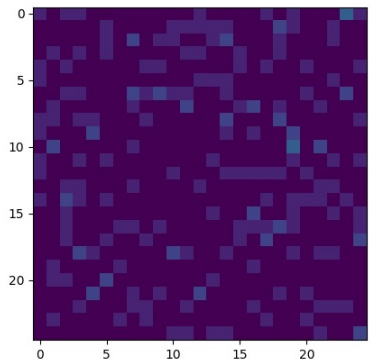
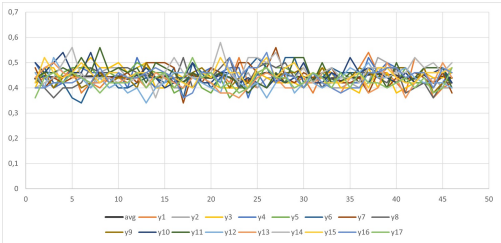
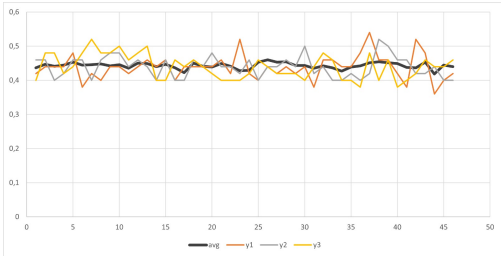


Figure 3.3: Smaller simulation, with a grid size of 25x25 and a population of 200.

Because all the code is properly annotated, it is easy to download it from section 5.1 and run it with different parameters. Seeing how the grid size or population count differs gives a visualization to the more "abstract" use of the β value. Now, for measles' simulation, the β value is extremely high (as it is an extremely contagious virus), and so it is run with 400 individuals in a 30×30 grid. The approximation for the β value can be seen below, in figure 3.4 .



(a) 17 Simulations



(b) 3 Simulations and average

Figure 3.4: Here we can see, to the left, 17 simulations' β lists in graph form, and, to the right, a simplified graphic, with only 3 simulations, and an average listed. The y-axis is β and x-axis are frames. We can see how with parameters $N = 30$, $M = 400$ the β value approaches $0.4 - 0.5$, which is Measles' disease transmission rate, as calculated below.

We aim for this β value because it originates from the following computation:

$$\beta = 12 \times \gamma$$

Where $R_0 = \beta/\gamma$ and $R_0 = 12 - 18$ [Met+17] and $\gamma = 1/25$, because we use a measles average for 25 days of infection (which is sligthly higher than the actual 21 days, but is compensated by a slightly lower β in the stochastic simulations).

3.2 SEIR-V-based stochastic simulations

We now introduce more parameters to the simulation:

- Population Count (M)
- Grid Size (N)
- Frames per day (FpD)
- Initially Exposed (exp_ini)
- Initially Infected (inf_ini)
- Initially Recovered (rec_ini)
- Initially Vaccinated (vac_ini)
- Transmissibility (τ)

Initially susceptible is calculated via subtracting from M all the other groups. Here, we can already see how the SEIR-V model looks like visually. In figure 3.5 we can already see the differences between this simulation and simulation 0.

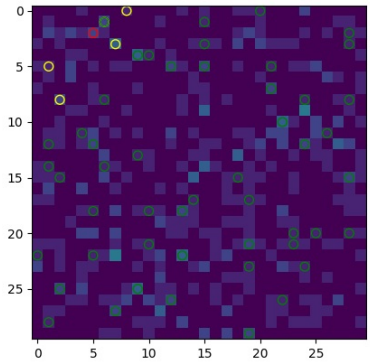
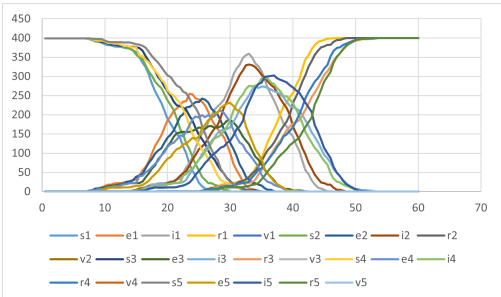


Figure 3.5: A single frame of simulation 1, with exp_ini = 1, vac_ini = 50, and $\tau = 0.9$. The other values are like simulation 0 or directly null. The frame shown is one after some days, where we can clearly see some red circles (Infected, the disease has already reached contagious rate), green circles (vaccinated or recovered, but they are all vaccinated here, since not enough time has passed for patient 0 to recover yet) and yellow circles (exposed, those who will begin presenting symptoms soon). The actors without a circle are susceptible.

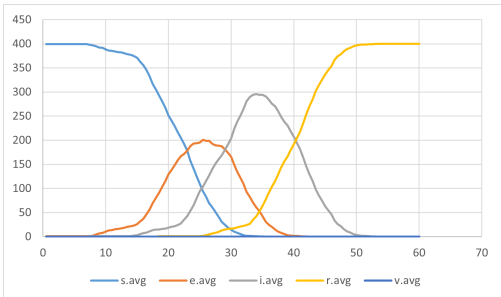
The code outputs all of the information to a .txt file, which we have compiled into graphs.

3.2.1 0% Vaccination coverage - Base case

Let us visualize how 0% vaccination rate affects measles, as a form of "base case", which we can also see in this video [3].



(a) 5 Simulations on 0% vaccination



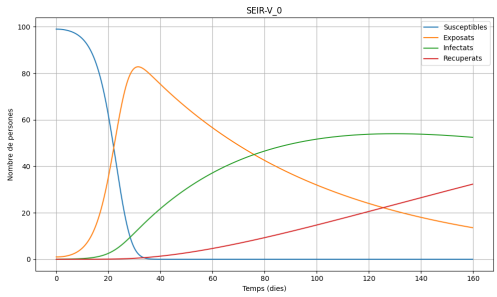
(b) Average of simulations on 0% vaccination

Figure 3.6: Here, the S,E,I,R,V indexes represent what one might expect, with the numbering indicating its parent simulation; the x-axis is population and the y-axis days. We can see in both graphs that, in the end, every person has contracted the illness.

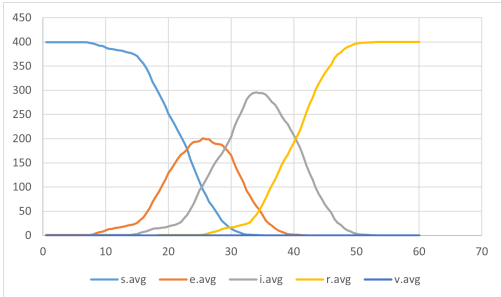
We can clearly see that measles ravages this population. This is a good example of how a densely

[3] If it is seen in a printed format, the link is: https://youtu.be/zbz7Gid0Vg0

packed city without any vaccine might fare against measles, and is reflective of historic examples, like this article which models almost 100 years of measles dynamics in London [Bec+19]. Let us compare it with what our SEIR-V model predicts, which uses a β computed from the following values:



(a) SEIR-V Simulation on 0% vaccination



(b) Stochastic Simulation on 0% vaccination

Figure 3.7: Axes are, again, population on the y-axis and days on the x-axis. The graph on the right is the average from figure 3.6, while the graph on the left is the one produced from the code running the differential equations.

We can see that the two figures match quite perfectly:

- In day 22, exposed goes above susceptible.
- In day 25, infected goes above susceptible.
- In day 32, there is no one who has not contracted the illness or is currently exposed/infected.

This proves that visual stochastic simulations which are actor-based can accurately follow predictions made with compartmentalized deterministic differential equations, at least up to this level of complexity. After everyone is infected, the differential equations fail to correctly predict the quick lowering of exposure and infection (which, logically, should take at most 7 and 7+18 days respectively after all susceptibles are infected if the exposure period is 7 days and infection period is 18), while the stochastic model correctly represents how the illness is a "burst" and dies off quickly - obviously, because it lacks others to infect, which doesn't happen in real-life properly connected communities.

It is estimated that, "during 2000-2023, an estimated 60.3 million measles deaths where averted by vaccination" [Min+24] . These results paint a grim picture of what a world without vaccination might look like, so let us step up the medical technology on our small population of 400 people.

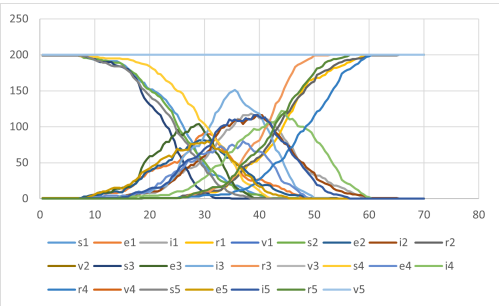
3.2.2 50% Vaccination Rate - Introducing the MMR vaccine

In an attempt to remedy the previous results, our population has decided to start selling vaccines. This means that some families will vaccinate, while others won't due to the cost. It also means that, even in rich families, there are those who can not vaccinate, since some are too vulnerable even for vaccination. Would a 50% vaccination rate properly cripple the disease, protecting

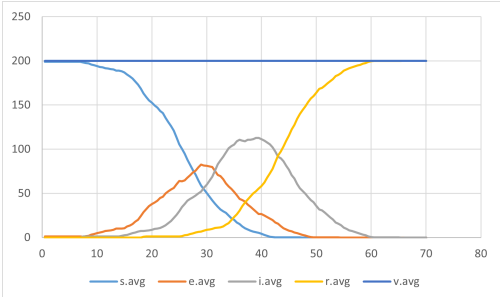
those not vaccinated? Logic tells us that, if half of the encounters an infected person has are with immune individuals, measles has half the chances to spread.

What we will be doing to properly simulate this is add a fixed amount of people to a "Vaccinated" group. In other systems, simulating different illnesses, one might want to account for outbreaks - appearance of new strains resistant to the vaccine - which might make the vaccine imperfect. However, measles' vaccine has an almost 100% lifelong effectiveness (97%-100% for two shots [Cen24]), so we will consider those vaccinated the same as those resistant - effectively, outside the system, decreasing contacts between infected-susceptible.

Here are the resulting graphs:



(a) 5 Simulations on 50% vaccination



(b) Average of simulations on 50% vaccination

Figure 3.8: Axes are, again, population on the y-axis and days on the x-axis. The graph on the left contains, again, 5 simulations, while the one on the right contains the average. The clear difference here is that there exists a constant, 200 actor strong value for vaccinated individuals, which doesn't participate in process of infection.

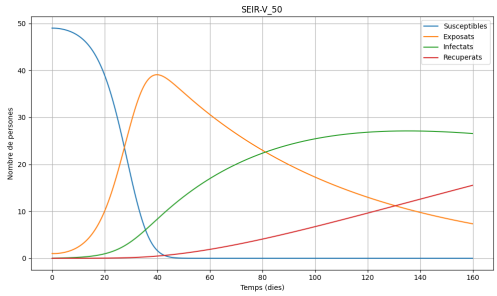
By selling vaccines, we have produced exactly what we wanted, which was to protect those with enough money to protect themselves. The vulnerable and poor are left behind both in these simulations and in real-life when we attempt to make a market out of life-saving drugs and treatment, which exemplifies why in most modern democracies vaccination is completely free.

It is important to note that the infection has been slowed:

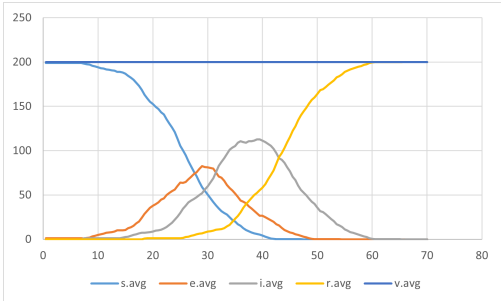
- In day 28, exposed goes above susceptible, compared to day 22 previously.
- In day 29, infected goes above susceptible, compared to day 25 previously.
- In day 41, there is no one who has not contracted the illness or is currently exposed/infected, compared to day 32 previously.

Note that the point where exposed and infected meet susceptible has gotten 2 days closer. While this model is stochastic, which implies that small random patterns like these are prone to happen, keep in mind this information in case it wasn't a random pattern.

Now, again, let us compare with the SEIR-V model:



(a) SEIR-V Simulation on 50% vaccination



(b) Stochastic Simulation on 50% vaccination

Figure 3.9: Axes are, again, population on the y-axis and days on the x-axis. The graph on the left is the SEIR-V prediction, while the one on the right contains the average shown in 3.8 .

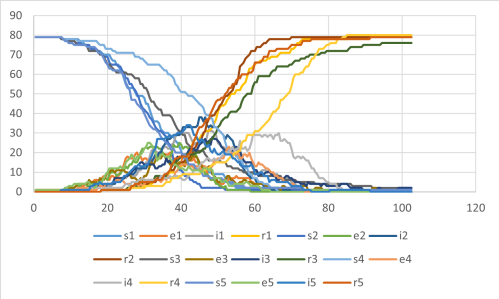
We will spare the reader the obvious comparisons already made before and establish again that the two graphics are almost identical up to decay in infected/exposed, which, again, SEIR-V doesn't model properly.

Here we don't see the pattern explained before as clearly, so it might be random. Again, though, for the sake of keeping the topic of a population fighting measles, let them notice this pattern too and attempt to make the infection process even slower by vaccinating everyone for free. After all, if the exposed never rise above the susceptible, there would be less infected per day, and the hypothetical hospitals might not be as crowded.

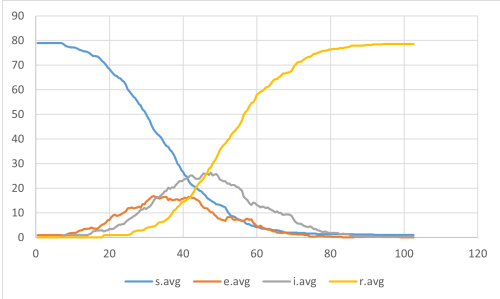
3.2.3 80% and 90% Vaccination Rate - Decreasing vaccination coverage

Here comes the crux of our argument. Let us say that our population, as we have already mentioned, implements free public vaccination. Let us also say, however, that a fraction of this population still does not wish to take the vaccine, maybe because of some unfounded fears certain social media circles might instil on them. Then, here are the results of their campaign, also in video form [4] :

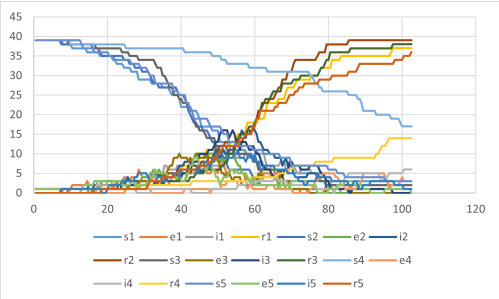
[4] https://youtu.be/JVuz6Jc3bW8



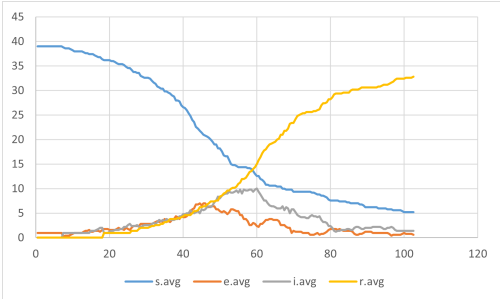
(a) 5 Simulations on 80% vaccination



(b) Average of simulations on 80% vaccination



(c) 5 Simulations on 90% vaccination



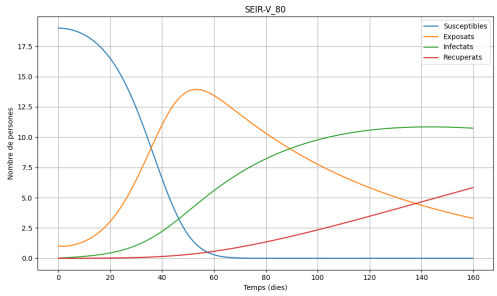
(d) Average of simulations on 90% vaccination

Figure 3.10: Axes are, again, population on the y-axis and days on the x-axis, although vaccinated has been removed for the sake of having a properly-scaled y-axis. We are now seeing how infected goes above susceptible in 80%, but exposed doesn't; and how not even infected goes above susceptible in 90%, which was correctly predicted by our conjecture.

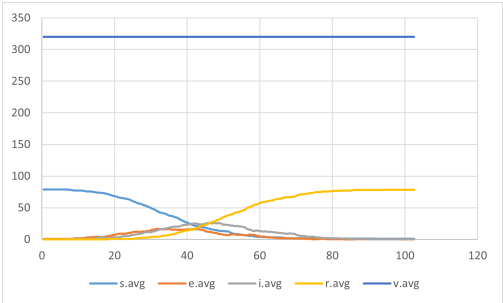
Our conjecture was correct. After vaccinating such a large portion of the population, the graph has flattened and most of our susceptible individuals have a place in the hospital. This was one of the main reasons for vaccinating during the Covid pandemic - aside from the obvious reason of protecting those vaccinated - since most of the crisis was permeated by headlines of full hospitals and 24-hour shifts from doctors [Bre20] .

However, notice the title of this section. These two simulations show not only what may improve in communities which recently implement free vaccination, but also what awaits those who privatize the vaccine or have large growing communities of anti-vax protesters. While the illness spreads slowly, still, even on 90% vaccination, more than half of the individuals who are not vaccinated have contracted the disease. It only requires an outbreak of 1 in 400 individuals for this to happen, showing how a single person could make more than half of the vulnerable in a small village have to go through measles. We also wish to remind the reader that 90% vaccination rate is uncomfortably close to the one the United States of America, the wealthiest nation on planet Earth [SDL24], currently has, as established in 1 . Some years it has even reached 90%, as can be seen in [WHO23].

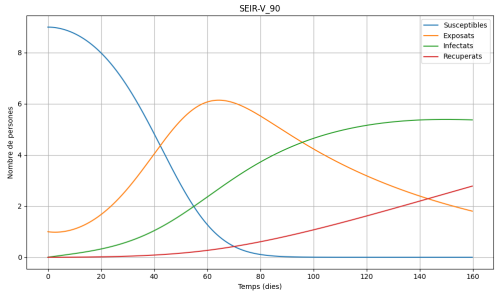
More relevant conclusions are drawn in 4 , but, for the time being, we show now the comparison with the SEIR-V differential equations.



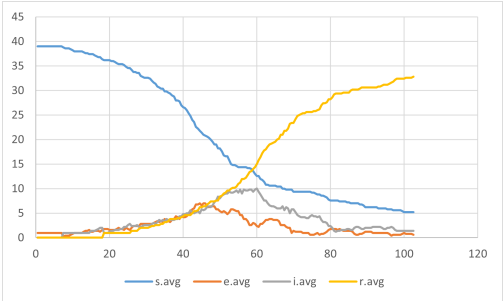
(a) SEIR-V Simulation on 80% vaccination



(b) Stochastic Simulation on 80% vaccination



(c) SEIR-V Simulation on 90% vaccination



(d) Stochastic Simulation on 90% vaccination

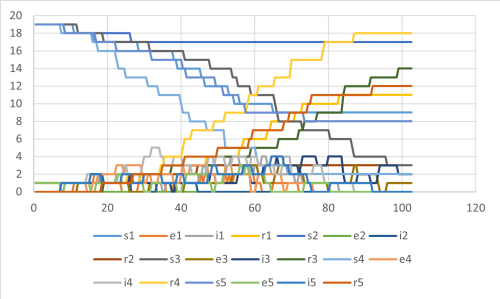
Figure 3.11: Axes are, again, population on the y-axis and days on the x-axis, and, again, vaccinated has been removed for the sake of having a properly-scaled y-axis. We now see that the SEIR-V equations do not match with the stochastic model any more.

These results stop being surprising when one realises that differential equations are not designed to model such small populations, and it is widely known that stochastic models more accurately represent these [SC19]. While for 400 individuals the SEIR-V equations predicted pretty well our stochastic model, they fail to consider the random chance that 1 in 10 individuals might meet another 5 suddenly, or that they might not meet any at all until they heal.

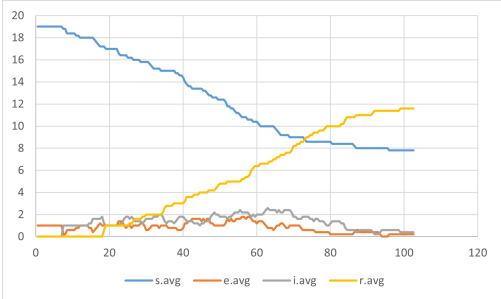
To quote the source already provided, "Consider for example a population of 20 centenarians, each having a probability 0.7 of surviving another year. The average number of survivors will be $20 \times 0.7 = 14$. However, due to the small size of the population, there will be random variations, and a probabilistic description of the population at the end of the year would be preferable." [SC19].

3.2.4 95% Vaccination Rate - Herd immunity

At this point, our models begin to break. To paint a proper picture, our stochastic model has 400 individuals; 5% of 400 is only 20 actors, out of which 1 is already infected. This represents an initial outbreak of a 5%, which is too high, for 20 people, which is too low a sample-size. Also, the time frame has gotten so big that the models end without ever reaching any conclusions. Nevertheless, here are the results:



(a) 5 Simulations on 95% vaccination

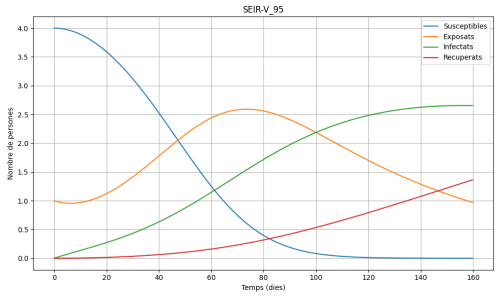


(b) Average simulation on 95% vaccination

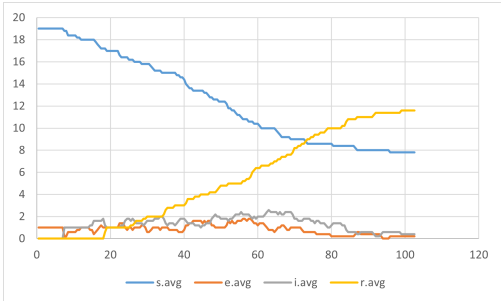
Figure 3.12: Axes are, again, population on the y-axis and days on the x-axis, although vaccinated has been removed for the sake of having a properly-scaled y-axis. There is a new trend of the ill and exposed lowering before the susceptible have been completely infected, which hints at what herd immunity implies - a quicker recovery of infected people than appearance of new infections. We can also see two new problems with the model here - the individual simulations are too blocky, and they never reach a conclusion.

Many studies have been done on herd immunity, and it is known that it begins sometime after 95%. Even though our simulation still predicts some chaos, it properly shows that the illness will probably die out before infecting every susceptible individual, which is a good sign. One must realise that in real-life, if someone contracts measles when the vaccination coverage is 95%, they will probably not meet other susceptible individuals who could not take the vaccine, because there is a strong network of others to help and protect them. Usual methods of transmission are also not that common in susceptible individuals any more, as babies and elders who aren't vaccinated don't usually take public transport (or move a lot in general), prepare food for others, work in teams or go to class.

As such, these simulations are no longer precise, but we leave the usual comparison here for the sake of completion:



(a) SEIR-V Simulation on 95% vaccination



(b) Stochastic Simulation on 95% vaccination

Figure 3.13: Axes are, again, population on the y-axis and days on the x-axis, and, again, vaccinated has been removed for the sake of having a properly-scaled y-axis. We now see that the SEIR-V equations still predict total infection, while it is highly unlikely. However, the stochastic simulation is not completely accurate either.

4 Conclusions

The testing using the SEIR-V model and simulation has resulted in data which provides some interesting insight into the development of measles in different contexts and epidemiologic simulation alike. The initial simulation showed how the constants used in the SEIR-V equations were indeed constant, despite minor fluctuations due to the stochastic nature of the project's methodology, which implies that visual stochastic simulations might be a proper alternative for scientific papers and divulgation journals alike - after all, someone outside the field of study would understand dots moving around much easier than some abstract differential equations or graphs produced from them.

Note also, however, the results in these simulations. As seen in simulation 3.2.1, one with no vaccination coverage at all, every susceptible individual is eventually infected, and the entire population ends up recovered within 60 days. This would mean that in a real-life scenario every person would be subjected to the illness, and have to roll the dice not to get the previously already mentioned 1 in 3000 chance of death.

Moving on to higher vaccination rates (80, 90 and 95%), we of course see how the whole evolution of the model involves less and less individuals, and how progression gets slower with the increasing vaccination rate, taking up to 100 days to have the small fraction of non-vaccinated individuals finally become recovered. While this is good, we are still seeing the smaller group be infected. Because people who can not take the vaccine are a constant number in all of these examples, they are part of a growing share of the infected portion of each simulation, and yet they still have to contract the illness at some point in their life. We must also consider the human loss of not only death, but weeks of a painful illness which makes people unable to socialize, study, or enjoy their day-to-day; and, if one were to acknowledge all of these loses, they may also be able to talk about the *economic* loss of having millions of workers stuck in bed.

To conclude, we have observed that the simulations show how higher vaccination rates lead to slower and considerably less impactful effects of a contagious disease. It can be inferred from the data that high enough vaccination percentages would turn potential population scale outbreaks

into particular cases with little to no transmission of measles, resulting in herd immunity, as we see in most of the modern world. However, the opposite extrapolation shows that a decrease in vaccination rates will eventually cause any particular case to become a potential start to a large-scale outbreak. Therefore, decreasing vaccination coverage must be avoided in order to preserve the herd immunity.

5 Annex

5.1 Code

While discussion of this code is outside the scope of this paper, it is necessary to establish that it was made exclusively by the authors for these results, and as such has required the greatest investment of time. Therefore, if the reader wants, they could download it from here [5] and play around with some parameters. SIM0 is for Simulation 0, SIM1 for Simulation 1, and SEIRV is the code that runs the differential equations. If any one of the simulations is laggy or outright wrong try to raise the time between updates at the bottom of the code (it will account for this itself and the days will be the same, but the simulations will run for less time; there is an annotation for the time it will run if one wishes to modify this too).

5.2 Data

This code has produced an extreme volume of data, much of it also extremely wrong. Nevertheless, one can access the data used for the graphs here [6] in sim-excels.zip or run the code themselves to acquire new, hopefully similar graphs.

[5] If it is seen in a printed format, the link is: https://github.com/Placeholder33/T07_E.git
[6] If it is seen in a printed format, the link is: https://github.com/Placeholder33/T07_E.git

References

[Ari+25] P. Ariyaratne et al. "Application and significance of SIRVB model in analyzing COVID-19 dynamics". In: Sci Rep 15, 8526 (2025). URL: https://doi.org/10.1038/s41598-025-90260-4.

[Bec+19] Alexander D. Becker et al. "Long-term dynamics of measles in London: Titrating the impact of wars, the 1918 pandemic, and vaccination". In: *PLOS Computational Biology* 15.9 (Sept. 2019), pp. 1–14. DOI: 10.1371/journal.pcbi.1007305. URL: https://doi.org/10.1371/journal.pcbi.1007305.

[Bon02] Eric Bonabeau. "Agent-based modeling: Methods and techniques for simulating human systems". In: *Proceedings of the National Academy of Sciences* 99.suppl_3 (2002), pp. 7280–7287. DOI: 10.1073/pnas.082080899. eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.082080899. URL: https://www.pnas.org/doi/abs/10.1073/pnas.082080899.

[Bre20] BreakingNews.ie. *Doctors working 24-hour shifts during Covid-19*. Accessed: 2025-05-04. 2020. URL: https://www.breakingnews.ie/ireland/doctors-working-24-hour-shifts-during-covid-19-1012197.html.

[Cen24] Centers for Disease Control and Prevention. *Measles Vaccination: What Everyone Should Know*. Accessed: 2025-05-04. 2024. URL: https://www.cdc.gov/measles/hcp/vaccine-considerations/index.html.

[IHM24] IHME, Global Burden of Disease. "Deaths due to Measles - with minor processing by Our World in Data". In: (2024). URL: https://ourworldindata.org/grapher/deaths-due-to-measles-gbd (visited on 04/27/2025).

[Met+17] C Jessica E Metcalf et al. "Estimating the reproduction number of measles from data on its impact: a systematic review". In: *The Lancet Infectious Diseases* 17.12 (2017), e420–e428. DOI: 10.1016/S1473-3099(17)30307-9. URL: https://www.sciencedirect.com/science/article/abs/pii/S1473309917303079.

[Min+24] Anna A Minta et al. "Progress Toward Measles Elimination - Worldwide, 2000-2023". en. In: *MMWR Morb Mortal Wkly Rep* 73.45 (Nov. 2024), pp. 1036–1042.

[RWC20] Anca Rădulescu, Cassandra Williams, and Kieran Cavanagh. "Management strategies in a SEIR-type model of COVID 19 community spread". In: *Scientific Reports* 10.1 (Dec. 2020), p. 21256. ISSN: 2045-2322. DOI: 10.1038/s41598-020-77628-4. URL: https://doi.org/10.1038/s41598-020-77628-4.

[SC19] Arni S.R. Srinivasa Rao and James R. Carey. "Chapter 11 - Behavior of Stationary Population Identity in Two-Dimensions: Age and Proportion of Population Truncated in Follow-up". In: *Integrated Population Biology and Modeling, Part B*. Ed. by Arni S.R. Srinivasa Rao and C.R. Rao. Vol. 40. Handbook of Statistics. Elsevier, 2019, pp. 487–500. DOI: https://doi.org/

10.1016/bs.host.2018.12.001. URL: https://www.sciencedirect.com/science/article/pii/S0169716118301056.

[SDL24] Anthony Shorrocks, James Davies, and Rodrigo Lluberas. *Global Wealth Report 2024*. Tech. rep. Accessed: 2025-05-04. UBS Group AG, 2024. URL: https://www.ubs.com/us/en/wealth-management/insights/global-wealth-report.html.

[Tan21] Linh Tang. *COVID-19 Modeling*. https://github.com/LinhTangTD/COVID19_Modeling. Accessed: 2025-05-04. 2021. URL: https://github.com/LinhTangTD/COVID19_Modeling.

[WHO23] World Health Organization. "Measles vaccination coverage". In: (2023). URL: https://immunizationdata.who.int/global/wiise-detail-page/measles-vaccination-coverage?CODE=USA&ANTIGEN=MCV1&YEAR=.

[WHO24] World Health Organization. "Measles - number of reported case". In: (2024). URL: https://www.who.int/data/gho/data/indicators/indicator-details/GHO/measles---number-of-reported-cases.